Web Design - Computer lab 5

Ing. Juraj Lazúr

Brno University of Technology, Faculty of Information Technology
Božetěchova 1/2. 612 66 Brno - Královo Pole

ilazur@fit.vut.cz

BRNO UNIVERSITY OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY

Where points

Published assignment 2. project

- **Deadline: 3. 5. 23:59**
- The subject of the project is the creation of a custom profile web presentation.

Motivation

- **64.35%** from mobile devices vs **35.96%** from desktops (Source).

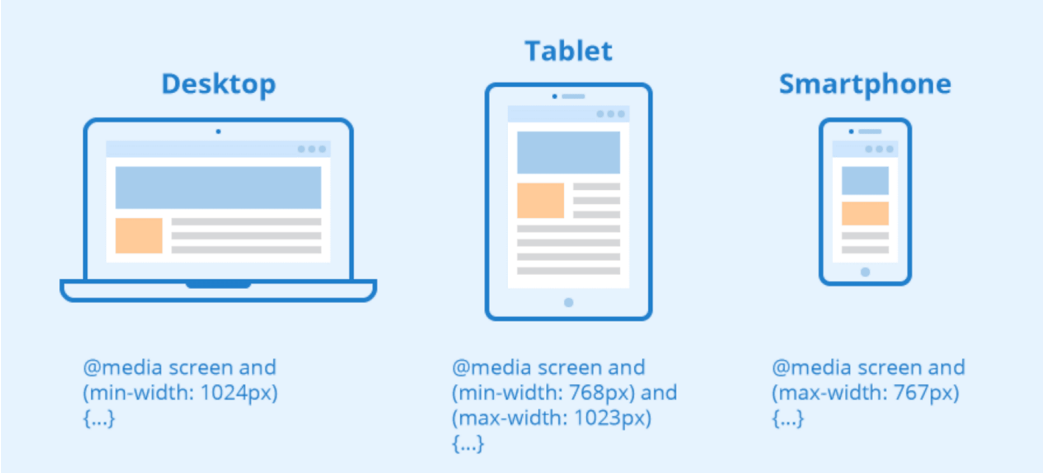

Image source: jeremymcgilvrey.com

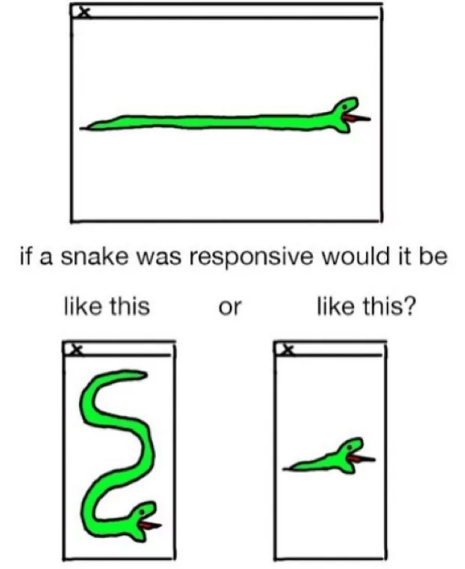

Image source: dev.to

Task 1: Viewport

1. Set up viewport tag:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- **The viewport** is the user's visible area of a web page.
- It "pretends" that your screen has **fewer pixels** than it actually does.
- The viewport is counted in **CSS pixels**, not real hardware pixels.
- Physical resolution is 1170×2532 pixels, but CSS viewport might be: 390×844
- This decouples layout width from physical resolution (DIPs) so fonts stay readable.

Task 2: Media Queries

2. Set up page responsiveness modes:

```
/* Default: Mobile View */
.container { width: 100%; }

/* Tablet & Desktop: Screen wider than 768px */
@media screen and (min-width: 768px) {
  .container { width: 80%; margin: 0 auto; }
}
```

- Use **Mobile First** approach (style the phone first).
- `min-width` triggers styles as the screen gets **wider**.
- **Breakpoints** allow the layout to "snap" into new positions.

Tip: Use **DevTools (F12)** to test these live!

- **xs:** $\leq$ 576px
- **sm:** $\geq$ 576px
- **md:** $\geq$ 768px
- **lg:** $\geq$ 992px
- **xl:** $\geq$ 1200px

Best Practices: Fluid Layouts

1. **Responsive Images**: Prevent images from "breaking" your layout on small screens:

```
img, video, canvas {
    max-width: 100%;
    height: auto;
}
```

2. **Use Relative Units**: Avoid using `px` for containers. Use `%` or `vw` for width, and `rem` for typography:

```
.container { width: 90%; max-width: 1200px; }
h1 { font-size: 2.5rem; }
```

What will be evaluated?

1. The whole site is **center-aligned**.
2. The whole site is readable at **different resolutions**.
3. **Validate** your solution (zero warnings/errors) using:
   - HTML validator
   - CSS validator